

CaseSense: Mapping the Legal Case Landscape with Multi-Dimensional Visual Exploration

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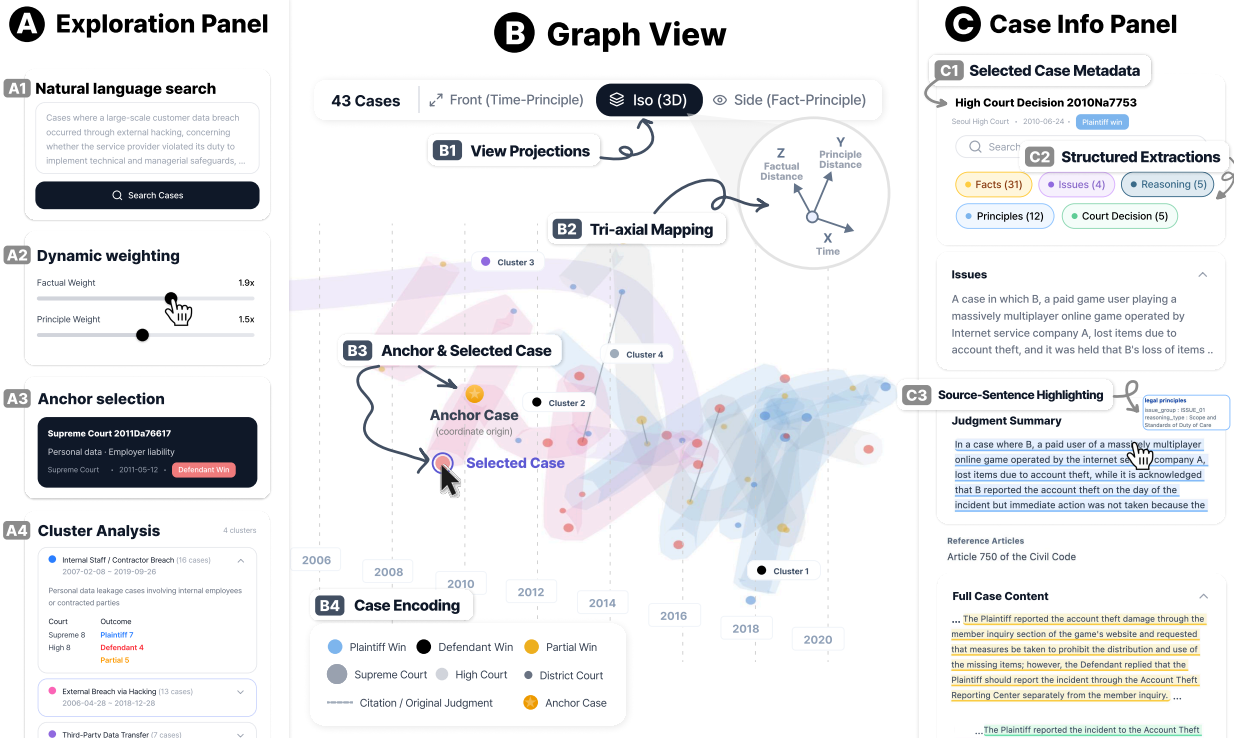


Figure 1: CaseSense overview. (A) The Exploration Panel supports natural language search (A1), dynamic axis weighting (A2), anchor selection (A3), and cluster-based thematic navigation (A4). (B) The Graph View places cases in a tri-axial space of time, principle distance, and factual distance (B2) with three purpose-constrained projections (B1), anchor and selection annotations (B3), and visual encoding of court level, outcome, and citations (B4). (C) The Case Information Panel shows selected-case metadata (C1) and structured extractions (C2) with source-sentence highlighting for verification (C3).

Abstract

Legal case research requires combining facts, legal principles, and timelines into persuasive arguments. Yet existing retrieval tools return one-dimensional ranked lists, collapsing this structure into a single relevance score. A formative study with five lawyers confirmed this flat output leads to missed cases, temporal gaps, and fragmented reasoning. We present **CaseSense**, a visual analytics system for multi-dimensional exploration via tri-axial mapping with dynamic weighting, anchor-based navigation, and purpose-constrained view projections. A within-subjects study with 16 legal professionals compared CaseSense against conventional workflows. CaseSense achieved complete key-case discovery in the Fact Synthesis task, with the largest gains in organization and connection identification. Participants adopted task-matched interaction strategies without instructions. These findings suggest encoding domain reasoning dimensions as manipulable spatial axes supports expert sensemaking, while separating spatial organization from AI-generated content preserves verification behavior.

CCS Concepts

• **Human-centered computing** → **Visualization application domains**; **HCI design and evaluation methods**.

Keywords

Legal Case Sensemaking; Multi-Dimensional Case Exploration; Visual Analytics for Law; Interactive Information Retrieval

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1 Introduction

Legal case research requires practitioners to combine factual circumstances, legal principles, and temporal evolution across dozens of cases [30, 50, 53]. This process is multi-dimensional: practitioners shift between factual and principle frames, trace how rulings evolve over time, and map inter-case relationships as their understanding develops [31]. Yet from keyword databases [45] through neural retrieval [9, 63, 69] to retrieval-augmented generation [13, 38], available tools typically return a one-dimensional ranked list that collapses these dimensions into a single relevance score. Even LLM-based systems that accept natural language queries preserve none of the dimensional structure practitioners reason along.

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Interactive visualization research has made high-dimensional spaces navigable through axis specification [27, 32], anchor-based navigation [14, 23], and coordinated views [11, 12], while LLM-augmented systems have shown that AI-generated structures can serve as substrates for human exploration [33, 61, 62]. These techniques, however, have seen little application to legal document collections [18], and where legal visualization has been attempted [51], it has been confined to citation topology rather than the reasoning dimensions practitioners use.

To understand this gap, we conducted a formative study with five Korean legal professionals. Three structural gaps emerged: keyword search forces practitioners to express nuanced questions through surface-level term matching (**Gap 1**); retrieval returns items without exposing their analytical structure (**Gap 2**); and static views cannot support dynamic perspective shifts (**Gap 3**). These gaps share a common root: current systems treat retrieval and exploration as separate stages, so the semantic signals that organize search results are discarded before practitioners can reason with them. This observation motivated three design requirements: support fact- and principle-driven search (**DR1**), place cases in multi-dimensional space (**DR2**), and enable perspective-shifting exploration (**DR3**).

In response, we developed **CaseSense**, a visual analytics system that maps the legal case landscape for multi-dimensional exploration of legal cases, built around three interaction mechanisms. Tri-axial mapping positions cases along decision date, principle distance, and factual distance, where the latter two are computed directly from the retrieval signals so that the dimensions organizing search become visible and adjustable through dynamic weighting. Anchor-based navigation redefines the coordinate origin around a user-selected case, recomputing all distances from the new reference point and shifting the retrieval-derived topology rather than merely changing a camera angle. Purpose-constrained view projections constrain each of the three views to a specific analytical purpose, channeling attention while preserving access to the full space; selection, anchor, and weight state persist across view switches. AI produces the structured extractions, cluster labels, and spatial coordinates that populate this space; the final sensemaking remains with the practitioner.

We evaluated CaseSense in a within-subjects study with 16 legal professionals across two tasks: Fact Synthesis (comparing factual circumstances across cases with divergent outcomes) and Principle Tracking (tracing legal-principle evolution across court levels and time periods). CaseSense improved sensemaking across both tasks, with the largest gains in organization and connection identification. Satisfaction and intention to use rose while perceived AI reliability did not change. Key-case discovery improved from 68.8% to 100% in Fact Synthesis and from 43.8% to 87.5% in Principle Tracking, with participants exploring more cases and reporting greater confidence in coverage. Each interaction mechanism contributed differently by task: participants spontaneously foregrounded the facts axis in Fact Synthesis and the principle axis in Principle Tracking, converged on task-matched views without instruction, and shifted from linear keyword search to iterative sensemaking loops.

Our contributions are threefold:

- **A retrieval-grounded interaction design** in which the factual–principle decomposition that structures retrieval simultaneously constitutes the axes of tri-axial mapping with dynamic weighting, anchor-based coordinate redefinition, and state-preserving, purpose-constrained view projections.
- **Empirical evidence** of gains in discovery completeness and sensemaking, with a discovery-trigger methodology attributing effects to individual mechanisms.
- **Design implications** for expert sensemaking systems: encoding domain reasoning dimensions as manipulable spatial axes, using anchor-based navigation for perspective shifting, and positioning AI as a structuring input rather than a content endpoint.

2 Related Work

We situate CaseSense at the intersection of three research areas: legal case retrieval, dimensionality manipulation and visual exploration, and LLM-augmented interactive sensemaking systems.

2.1 Legal Case Retrieval

Legal case retrieval has evolved from Boolean keyword matching [45] through probabilistic ranking [52] to neural approaches, yet the output format—one-dimensional ranked lists—has remained unchanged. On the neural front, domain-specific pretraining [9, 10], structure-aware models [39, 57], dense retrieval [17, 69], and graph-based citation propagation [63, 64] have improved semantic matching, while hybrid lexical-dense pipelines [3] outperform either alone. A complementary challenge is that LLMs hallucinate legal knowledge [13, 44]; retrieval-augmented generation (RAG) addresses this by grounding generation in verifiable sources [4, 38], with legal-domain adaptations targeting terminology mismatch [49] and structural analogy [68]. Both directions are active in the Korean legal context: retrieval benchmarks have advanced semantic matching for judgment prediction and case search [22, 26, 29], and RAG has been applied to statutory question answering [54, 56].

Despite these gains in retrieval accuracy, all of these systems collapse the multi-dimensional structure of legal cases—factual circumstances, legal principles, temporal evolution, and citation relationships—into a single relevance score. CaseSense addresses this gap by preserving three semantically meaningful dimensions as explicit, manipulable spatial axes so that practitioners can explore how cases relate, rather than improving what is retrieved.

2.2 Dimensionality Manipulation and Interactive Visual Exploration

Dimensionality reduction techniques (e.g., t-SNE [66], UMAP [46]) produce navigable layouts but sacrifice axis interpretability [16, 60]. Interactive axis specification [27, 32] and concept-aligned latent dimensions [42, 67] have restored legibility, yet axes remain domain-agnostic. To give users further control over layout structure, slider-based dynamic queries [1, 58], weighted multidimensional scaling [15, 55], and semantic interaction [14] let users reshape projections interactively. Anchor-based approaches complement such weighting by repositioning data relative to a reference point—either

through control-point projections [23] or observation-level interactions [14, 27]—thereby shifting the analytical frame without modifying axis definitions. Chronotome [40] similarly encodes time along the Z-axis with Front/Iso/Side preset views, though its axes reflect force-directed layout rather than domain-semantic dimensions. When no single projection can answer all analytical questions, multiple coordinated views distribute the exploration across complementary perspectives [11, 12].

These techniques, however, have seen limited application to legal document collections. A recent survey [18] documents that legal-domain visual analytics remain largely unexplored. Where visualization has been attempted, it has been confined to citation topology [51] or to statutory amendment tracking rather than case law [28]. CaseSense fills this gap through a structural unification: the same factual–principle decomposition that drives retrieval simultaneously defines the semantic axes practitioners manipulate, so that retrieval topology becomes more navigable as a coordinate system rather than collapsed into a ranked list.

2.3 LLM-Augmented Interactive Sensemaking Systems

LLMs have been integrated into interactive sensemaking through two broad strategies. The first positions the LLM as a conversational partner: spatial canvases enable multilevel abstraction [62], structured overviews scaffold cold-start sensemaking [41], and multi-agent systems orchestrate multiple perspectives [24, 48]. In these systems, however, LLM outputs serve as conversational artifacts to browse rather than structured substrates to navigate through domain-specific techniques. A second strategy addresses this by using the LLM as a generator of structured representations that feed into purpose-built interfaces: LLMs populate design spaces [61], induce concept-scored dimensions [33], extract hypergraph entities [36], and generate insight annotations [19, 70].

However, neither strategy has been applied to legal case spaces, where AI outputs must align with the reasoning dimensions practitioners already use. CaseSense follows the second approach: LLMs extract legal elements and generate tri-axial coordinates as inputs to the visual interaction space, enabling practitioners to navigate the legal case space through the very reasoning dimensions that structure legal argumentation itself.

3 Formative Study

To identify where existing tools fail to support legal sensemaking—even with LLM assistance—we conducted a formative study with five Korean legal professionals. Three structural gaps emerged, each motivating a design requirement for CaseSense.

3.1 Participants and Procedure

We recruited five participants through professional legal communities (Table 1). Each participated in a 60-minute session consisting of three phases: (1) a workflow walkthrough in which participants demonstrated their current research process with their preferred tools; (2) a critical incident interview in which they described recent cases where research failed or produced unexpected outcomes; and (3) a speculative design session in which they reacted to wireframes of multi-dimensional visualizations. Sessions were conducted in

Table 1: Formative study participant demographics.

ID	Gender	Age Range	Profession	Specialty	Exp.	Case DB	AI Tools
F1	F	25–34	Law firm associate	Contract, Tort	5 yrs	CaseNote ^a , Lawnb ^b	ChatGPT ^c , Perplexity ^f
F2	M	25–34	In-house counsel	Data protection	4 yrs	Lbox ^c , CaseNote ^a	ChatGPT ^c
F3	F	35–44	Prosecutor	Property, Fraud	6 yrs	Lawnb ^b	None
F4	M	25–34	Law firm associate	Real estate	3 yrs	CaseNote ^a , Lbox ^c	ChatGPT ^c , Lbox AI ^c
F5	F	25–34	Judicial researcher	Liability, IP	5 yrs	Court internal DB ^d	ChatGPT ^c , Perplexity ^f

^a <https://casenote.kr> ^b <https://lawnb.com> ^c <https://lbox.kr> ^d Courts' internal system (not public) ^e <https://chat.openai.com> ^f <https://perplexity.ai>

Korean, audio-recorded, and transcribed. Two researchers independently coded transcripts using open coding, then consolidated themes through iterative discussion.

Open coding surfaced a workflow arc—issue framing, candidate retrieval through databases and AI tools, relevance triage, cross-case synthesis, and currency checking—and the three gaps below emerged as breakdowns at the retrieval, synthesis, and currency stages, grounding the design requirements in observed practice.

3.2 Findings and Design Requirements

Five Korean legal professionals identified three structural problems in current search tools, each motivating a design requirement for CaseSense.

3.2.1 Gap 1: Search Vocabulary Does Not Match Analytical Intent → DR1: Support Fact- and Principle-Driven Search. Keyword search forces practitioners to express nuanced legal questions through surface-level term matching. All five participants described this as the central frustration of their daily work. F3: *“Finding a case with matching facts is what matters most. But keyword search only matches terms. I can’t search for ‘the court ruled differently despite similar facts.’”* F1: *“It’s like touching an elephant blindfolded. I broaden keywords when results are thin, but I never know whether I’ve covered the whole animal.”* LLMs partially address this mismatch by accepting natural language queries, but they introduce a different problem: plausible-sounding yet unverifiable results. F5: *“ChatGPT gives a structured answer, but I still need to verify every case separately. It fabricates case numbers. It’s useful as a starting outline, but not as a search tool.”* Such fabrication is a known failure of ungrounded generation that retrieval grounding largely resolves [38, 44]; what grounding does not change is the output modality—results still arrive as linear text whose multi-dimensional structure practitioners must reassemble themselves, and this perceivability gap, not retrieval accuracy, is what CaseSense targets. This gap motivates a search pipeline that accepts natural language queries and decomposes them into factual and principle components, issuing each against a dedicated embedding index to retrieve cases that match both the situation at hand and the legal principle in question.

3.2.2 Gap 2: Retrieval Returns Items, Not Structure → DR2: Place Cases in Multi-Dimensional Space. Even when retrieval succeeds, the output (a ranked list) discards the analytical structure that practitioners need to build arguments. F2: *“AI tools give me a flat list of relevant cases. But what I actually need is to see how cases relate to each other: which ones build on others, which ones diverge. The AI knows the answer but hides the map.”* Through workflow

walkthroughs, we observed that practitioners reason along three primary dimensions when evaluating a legal case: temporal position, factual similarity, and legal-principle alignment. F4: *“When I find a relevant case, I immediately ask: When was it decided? Has the Supreme Court ruled since? Are the facts close enough? Which legal principle did it apply? Every case sits at the intersection of these questions.”* Yet every existing tool collapses these dimensions into a single relevance score. The consequence is professional risk, not merely inconvenience. F5: *“I once submitted a brief relying on a 2015 High Court ruling, not realizing the Supreme Court had overturned the underlying principle in 2018. It appeared at the top because keywords matched, but it was temporally obsolete.”* This gap motivates a tri-axial coordinate space structured by decision date, principle distance, and factual distance, with thematic clusters overlaid to surface meaningful groupings. A dynamic weighting mechanism allows users to adjust the relative emphasis of each axis, making the analytical structure not only visible but interactively adjustable.

3.2.3 Gap 3: Static Views Cannot Support Dynamic Reasoning → DR3: Enable Perspective-Shifting Exploration. Even when case structure is visible, practitioners must shift analytical perspective to reason effectively, yet existing tools lock users into a single fixed view. F5: *“I need to see both the timeline of how liability standards evolved AND which cases have similar factual patterns. No tool lets me see both at once. I have to mentally switch between chronological and thematic views.”* Korean civil law sharpens this challenge further: Supreme Court dismissals scatter substantive analysis across poorly linked lower-court decisions, forcing practitioners to synthesize rulings across court levels and time periods to trace how legal principles evolve. This gap motivates two interconnected mechanisms: anchor-based navigation (redefining the coordinate origin around a user-selected case to shift the entire spatial arrangement) and purpose-constrained view projections (Front for temporal-principle analysis, Side for factual comparison, and ISO for holistic overview).

These three design requirements motivate the system design introduced in §4: **DR1 (fact- and principle-driven search)**, **DR2 (multi-dimensional case placement)**, and **DR3 (perspective-shifting exploration)**.

4 CaseSense : Design and Implementation

CaseSense maps the retrieved case landscape into a three-dimensional space where the factual–principle decomposition that drives retrieval simultaneously defines the spatial axes users manipulate. This structural unification makes retrieval topology directly navigable: practitioners explore case relationships along factual distance, principle distance, and decision date, rather than scanning a

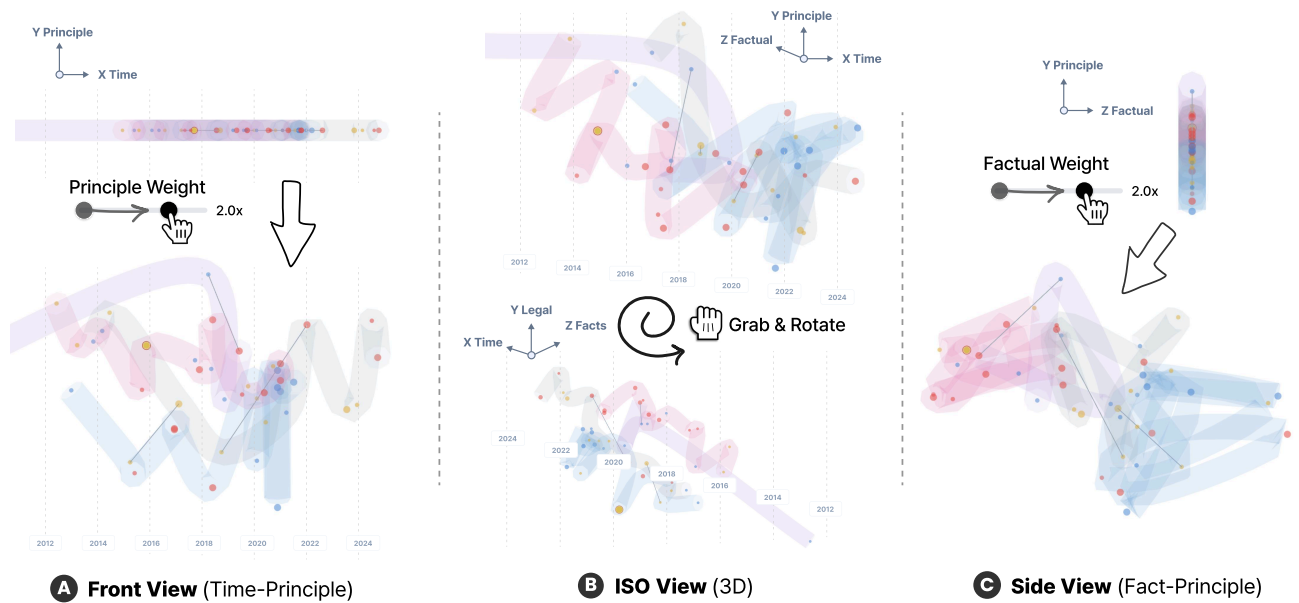


Figure 2: Purpose-constrained view projections and dynamic weighting. (A) Front view projects the time–principle plane with a principle weight slider. (B) ISO view preserves all three axes with free rotation. (C) Side view projects the facts–principle plane with a factual weight slider. Each projection suppresses one axis to focus attention on the two most relevant dimensions.

ranked list. The system addresses the design requirements identified in the formative study through three interaction mechanisms: (1) **tri-axial mapping with dynamic weighting**, (2) **anchor-based navigation**, and (3) **purpose-constrained view projections**, supported by a backend pipeline that produces the structured extractions and spatial coordinates.

4.1 User Interface

The CaseSense interface consists of three zones: a Graph View (center) for spatial exploration, an Exploration Panel (left) for thematic navigation, and a Case Information Panel (right) for detailed inspection. These panels are simultaneously visible and linked, forming coordinated multiple views in the conventional sense; the Front, Side, and ISO modes are state-preserving, purpose-constrained projections shown one at a time. A natural language search bar provides the primary entry point for exploration (Figure 1).

4.1.1 Tri-Axial Mapping with Dynamic Weighting (DR2). Cases are positioned along three semantically grounded axes: X encodes time relative to the anchor, Y encodes principle distance, and Z encodes factual distance. The anchor is always placed at the origin, so proximity reflects similarity and distance reflects divergence. Unlike dimensionality-reduction approaches (e.g., t-SNE, UMAP), each axis carries explicit legal meaning, making the dimensions practitioners already reason along directly readable from spatial position. A Dynamic Weighting panel lets users stretch or compress each axis in real time, foregrounding the dimension most relevant to their current question.

4.1.2 Anchor-Based Navigation (DR3). The default anchor is the top-ranked retrieved case. Practitioners can designate any other node as the new anchor through an explicit “Set as Anchor” action; this redefines the coordinate origin and recomputes all positions relative to the new reference point. Unlike fisheye distortion or semantic zoom, this constitutes a coordinate redefinition. The entire spatial arrangement shifts to reflect distances from the new anchor, enabling practitioners to re-examine the same case set from a different interpretive standpoint.

4.1.3 Purpose-Constrained View Projections (DR3). Three purpose-constrained views provide complementary analytical frames:

- **Front view** (time–principle plane): Suppresses the facts axis, optimized for tracking how legal–principle distance from the anchor evolves over time.
- **Side view** (facts–principle plane): Suppresses the time axis, optimized for comparing factual and principle distances regardless of when cases were decided.
- **ISO view** (free-rotation 3D overview): Preserves all three dimensions, enabling holistic overview and unconstrained exploration.

View switching is instantaneous and preserves selection state, anchor position, and weight settings. In the two constrained views, rotation is disabled to maintain axis interpretability; the ISO View allows free rotation. Each projection eliminates one dimension to focus analytical attention on the two most relevant to the current reasoning task, collectively addressing the need to shift between chronological and thematic perspectives identified in DR3.

4.1.4 Supporting Features (DR1, DR2). Natural Language Search (DR1) accepts free-form queries and decomposes them into factual and principle components to drive structured retrieval. Cluster Overview (DR2) encodes cluster membership as node color, as a translucent band that traces each cluster’s members along the time axis following spatial–temporal encoding [40], and as labeled, color-coded cards, enabling rapid thematic filtering. Because membership (from *Legal_Issue*) and position (anchor-relative distances) are defined in different spaces, clusters are intentionally non-compact in the fact–principle plane; the bands therefore convey temporal continuity and membership rather than spatial extent, and we avoid region encodings that presuppose compactness (e.g., convex hulls, bubble sets, or density contours), reading cluster composition, size, and time span from the cards instead. HDBSCAN noise points are excluded from the bands and appear under an explicit “Other” card. Case Information Panel (DR1) displays structured extractions of the selected case in collapsible sections, with sentence highlighting that links each extracted field to its source sentence in the original decision, enabling practitioners to verify AI-extracted fields against the judicial text.

4.1.5 Interaction Walkthrough. A typical session forms an iterative loop (§6.3): a query is decomposed and retrieved, and the Graph View renders candidates around the default anchor. The practitioner scans the cluster cards for orientation, switches to the task-matched projection (Front for principle evolution, Side for factual comparison) and adjusts its weight slider, inspects a promising node’s extractions and verifies them against source sentences, then re-anchors on it and repeats: search → visualize → anchor → weight → view → discover → refine.

4.2 Backend Architecture

All components use GPT-5.1 (gpt-5.1-2025-11-13) for all language generation tasks and OpenAI’s text-embedding-3-large for all vector representations. Vector indices are maintained in ChromaDB. Indexing operations are performed once prior to deployment; query-time operations execute per query (Figure 3).

4.2.1 Case Collection and Structured Field Extraction. CaseSense is built on approximately 50,000 Korean civil cases obtained through the Korean Law Information Service API [47], preprocessed to extract citation links and appeal connections across court levels. Temporal metadata is preserved as a first-class attribute, constituting the X-axis of the visualization coordinate system.

Raw cases, however, blend factual narrative, statutory citation, and judicial reasoning into undifferentiated prose. We apply LangExtract [20] to decompose each document into five verbatim-grounded fields: *Legal_Issue* (central contested question), *Facts* (factual circumstances), *Legal_Principle* (applicable statutes and principles), *Court_Reasoning* (logical chain from facts to outcome), and *Court_Decision* (final ruling). All extracted text must appear verbatim in the source document, preserving original judicial language and reducing semantic drift. The extraction prompt and output schema are documented in Appendix B.1.

4.2.2 Thematic Clustering and Vector Indexing. Thematic cluster labels are derived from the *Legal_Issue* field extracted above to give practitioners a coarse orientation within the case space before any

query is issued. All *Legal_Issue* texts are grouped via HDBSCAN [7]; for each resulting cluster, the LLM generates a concise noun phrase summarizing the shared legal theme (e.g., “External Data Breach via Hacking”). The resulting `cluster_id` and `cluster_label` are stored as metadata and applied at visualization time for color-coded node grouping without query-time computation.

Keeping this layer offline is deliberate: clusters serve as a stable, query-independent orientation map, while query-specific differences between same-cluster cases are expressed through per-query Y and Z positions; reclustering at query time would destabilize the map and forfeit the offline label validation (Appendix C). The cluster labeling prompt is detailed in Appendix B.2.

The multiple spans LangExtract returns per field are contiguous pieces of one field; for each case we concatenate them in document order and embed each field once, yielding one $\mathbf{v}_f(c)$ and one $\mathbf{v}_p(c)$ per case, so similarity in the layout denotes whole-field similarity.

The *Facts* and *Legal_Principle* fields are independently embedded and indexed in two ChromaDB collections—*facts_db* and *principle_db*—because factual similarity and principle similarity serve different roles that must remain separable. *facts_db* drives candidate selection via ANN search; *principle_db* provides the scoring signal that positions each retrieved case along the Y-axis. Merging them into a single index would conflate two analytically distinct signals and undermine both retrieval precision and spatial interpretability.

4.2.3 Query-Driven Candidate Selection. At query-time, the language model decomposes the natural language query into two components: a *facts_query* capturing factual circumstances, and a *principle_query* identifying the relevant legal principle. The query decomposition prompt is detailed in Appendix B.3.

Retrieval proceeds in two stages. In the first, *facts_query* is issued against *facts_db* via approximate nearest-neighbor (ANN) search, retaining only cases exceeding a factual similarity threshold ($s_f(c) > \tau_f$, empirically initialized to 0.6) as candidates C . This factual filter exploits the specificity of factual descriptions to constrain the candidate space efficiently. In the second, principle similarity $s_p(c) = \text{sim}(\text{principle_query}, \mathbf{v}_p(c))$ is computed for each $c \in C$ by fetching its vector from *principle_db*; no threshold is applied—cases with low principle similarity remain in the candidate set and are positioned farther from the anchor along the Y-axis.

4.2.4 Anchor Selection and Coordinate Projection. The *Anchor* is selected as the top-ranked candidate by combined score and placed at the origin (0, 0, 0). All candidates are then projected as:

$$\text{anchor} = \arg \max_{c \in C} \alpha \cdot s_f(c) + \beta \cdot s_p(c) \quad (\alpha = \beta = 0.5) \quad (1)$$

$$X(c) = \text{decision_date}(c) - \text{decision_date}(\text{anchor}) \quad (2)$$

$$Y(c) = 1 - \text{sim}(\mathbf{v}_p(c), \mathbf{v}_p(\text{anchor})) \quad (3)$$

$$Z(c) = 1 - \text{sim}(\mathbf{v}_f(c), \mathbf{v}_f(\text{anchor})) \quad (4)$$

Cases near the origin share close principle and factual characteristics with the anchor; cases farther out diverge along one or both dimensions. When the user selects a different anchor node, all coordinates are recomputed, constituting a full perspective shift across the visualization space.

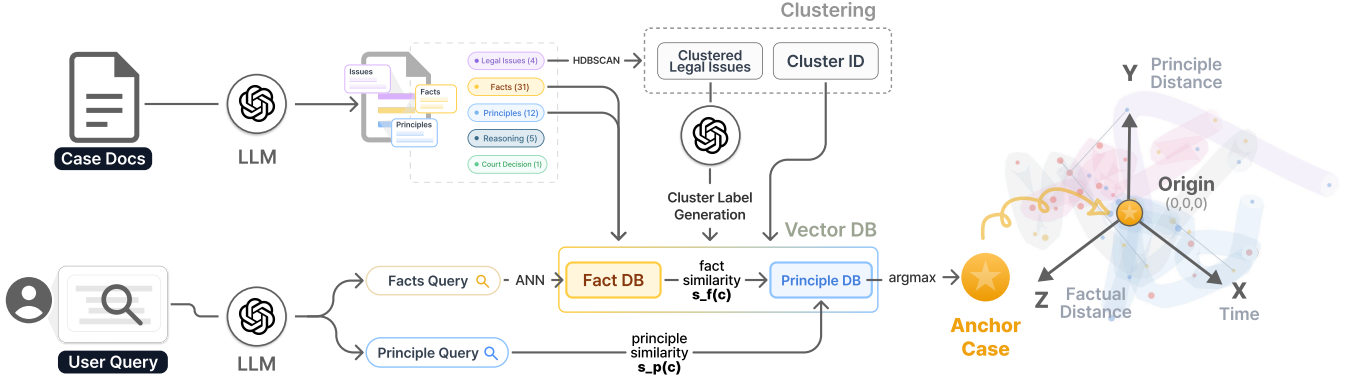


Figure 3: Backend pipeline of CaseSense.

4.3 Technical Evaluation

To validate the retrieval pipeline prior to the user study, we evaluated 40 legal research tasks across four civil law domains against ground truth established by F5 (judicial researcher, 5 years). Retrieval quality was assessed automatically, achieving Precision@5 of 0.82 ($SD=0.12$), Anchor Accuracy@1 of 0.875, and MRR of 0.88 ($SD=0.10$). Cluster coherence was rated by F4 and F5 independently, yielding 85.0% coherent clusters and 80.0% label accuracy (Fleiss' $\kappa=0.76$). Extraction quality was not benchmarked in isolation, which we note as future work; three factors nonetheless bound its impact: Precision@5 is computed on the extracted-field embeddings and thus serves as a coarse end-to-end check, the verbatim-span constraint biases errors toward omission rather than fabrication, and whole-field embeddings dilute a missed span rather than letting it alone fix a position—consistent with participants' use of the extractions as a reliable first-pass filter (§6.2). These results confirmed that no systematic pipeline errors would confound the user study findings. Details appear in Appendix C.

5 User Study

We conducted a within-subjects study with 16 legal professionals comparing CaseSense against conventional workflows across two legal research tasks.

- **RQ1 (Sensemaking):** Does CaseSense enhance sensemaking capabilities—specifically organization, connection identification, and insight generation?
- **RQ2 (Discovery):** Does CaseSense improve the completeness and accuracy of key-case discovery?
- **RQ3 (Interaction Mechanism Contributions):** How does each interaction mechanism contribute to discovery, and do contributions differ across task types?

5.1 Participants

We recruited $N=16$ legal professionals (10 female, 6 male; mean age 30.3, $SD=2.3$) through snowball sampling from law firms and professional legal communities. All had completed civil law training and regularly used case law databases, averaging 3.9 years of experience (range: 1–8); full demographics and tools used are in Table 4. With 16 participants, estimated power is 0.85 for a within-subjects

effect of $d_z=0.8$ (paired t -test, two-tailed $\alpha=.05$); we supplement significance testing with effect sizes and triangulate across behavioral logs, ratings, and interviews. Participants were compensated 150,000 KRW ($\approx \$100$ USD).

5.2 Conditions and Tasks

In the **Baseline**, participants used their conventional tools—commercial databases (CaseNote [8], Lbox [35], Lawnb [34]), search engines, and AI assistants (ChatGPT, Perplexity)—without restriction, preserving ecological validity. In the **CaseSense** condition, participants used the full system: three interaction mechanisms, Cluster Overview, and the Case Information Panel with structured extractions.

Two tasks, validated by a judicial researcher (F5) who defined key cases and evaluation criteria, represented distinct analytical demands. **Task 1 (Fact Synthesis)** required researching contract termination cases involving housing cooperatives, primarily demanding factual comparison across cases with divergent outcomes. **Task 2 (Principle Tracking)** required tracing the evolution of service providers' liability in data breach cases, primarily demanding temporal-principle reasoning across court levels and time periods. These tasks were chosen to test whether the three interaction mechanisms contribute differently by analytical demand.

5.3 Study Design and Procedure

The protocol was approved by our university's IRB. A balanced Latin square across four groups ensured each task appeared equally in each position and each condition appeared equally in the first and second session. Mann-Whitney U tests confirmed no significant ordering or task-sequence effects ($p>.05$ for all metrics).

Each 120-minute session comprised a 5-minute introduction, a 10-minute CaseSense tutorial, two task sessions (average 15–20 minutes each), post-task questionnaires after each task-condition pair, and a 30-minute semi-structured interview.

5.4 Measures and Analysis

5.4.1 Post-Task Questionnaire (RQ1). Thirteen items on 7-point Likert scales assessed two dimensions: six usability/workload items and seven sensemaking items (Table 6). Internal consistency was

acceptable (Cronbach's $\alpha=0.79$ for usability; $\alpha=0.83$ for sensemaking). We used Wilcoxon signed-rank tests (two-tailed) with Holm-Bonferroni correction applied within each family (6 usability, 7 sensemaking) per task. Effect sizes are matched-pairs rank-biserial correlations [25]. Paired t -tests as a robustness check reached identical conclusions for 86% of comparisons; the four disagreements all involved t -tests reaching significance where Wilcoxon did not, confirming our nonparametric results are conservative.

5.4.2 Performance Metrics (RQ2). For each session we recorded three *case events*, used consistently in Table 2 and Figure 6: *searched* (a unique case opened during exploration), *cited* (a case incorporated into the participant's task output—coded from outputs, think-aloud, and recordings, as the interface has no cite button), and binary *key-case discovery* (validated by F5). All performance metrics are reported descriptively (M , SD , counts, and rates) because the primary contrasts of interest, key-case discovery and exploration breadth, are binary or count variables for which descriptive comparison is more informative than null-hypothesis testing at $N=16$.

5.4.3 Behavioral Logs (RQ3). CaseSense logged every interaction with timestamps: search queries, case clicks, anchor changes, weight adjustments, view transitions, and cluster card clicks. Baseline sessions were screen-recorded and coded by four researchers. From these logs we derived discovery triggers: for each discovery event (a case click that led to citation or key-case identification), we inspected the preceding 4-event window and recorded all interaction types present. Discoveries whose 4-event windows contained no interaction mechanism use (only case clicks or search queries) were classified as direct exploration. We also computed view dwell time (proportion of session spent in each view). These metrics are reported descriptively.

5.4.4 Semi-Structured Interviews (RQ1–RQ3). A 30-minute interview explored reasoning strategies, feature preferences, and between-condition comparisons. Two researchers independently coded four transcripts to establish a codebook (inter-rater agreement Cohen's $\kappa=0.82$ on the initial four transcripts); the remaining 12 were divided for primary coding, generating 867 codes refined into themes through iterative discussion [5].

6 Findings

CaseSense improved sensemaking, key-case discovery, and elicited task-appropriate use of each interaction mechanism. We report converging evidence from behavioral logs, post-task ratings, and semi-structured interviews, organized by three research questions followed by participant-identified challenges.

6.1 RQ1: Relational Sensemaking Through Spatial Encoding

CaseSense improved sensemaking across both tasks, with consistent significance in Fact Synthesis and mixed effects in Principle Tracking. Usability gains followed the opposite pattern, concentrating in the more complex Tracking task (Figure 4, 5).

6.1.1 Strongest Gains in Organization and Connection Identification. Organization (i.e., the ability to classify, group, and structure cases) and connection identification showed the strongest and most

consistent effects. In Fact Synthesis, both reached maximum effect sizes ($r=1.000$; $p=.004^{**}$; $p=.006^{**}$), meaning every participant with a directional change favored CaseSense. Connection identification remained the strongest effect in Principle Tracking ($p=.008^{**}$, $r=1.000$), with organization also significant ($p=.043^{*}$, $r=.790$). The remaining Fact Synthesis items were all significant: structure finding ($p=.014^{*}$, $r=.895$), comprehension ($p=.048^{*}$, $r=.844$), contextual understanding ($p=.048^{*}$, $r=.724$), and gap filling ($p=.048^{*}$, $r=.583$). Principle Tracking showed more variable effects, with structure finding ($p=.078^{\dagger}$, $r=.588$), comprehension ($p=.078^{\dagger}$, $r=.642$), and gap filling ($p=.087^{\dagger}$, $r=.576$) at marginal levels.

P3: *"Before, I held six rulings in working memory: which overturns which, which pair shares facts but diverges. By the fourth case the mental map collapsed. With CaseSense those relationships are on screen, so my effort shifts from constructing the map to evaluating it."*

That Fact Synthesis yielded uniform significance while Principle Tracking showed more variable effects suggests that CaseSense's spatial encoding is immediately effective for factual comparison, but the added temporal complexity of tracking moderates gains on items that need deeper engagement.

6.1.2 A Shift from Sequential Filtering to Structural Reasoning. Beyond individual sensemaking items, CaseSense appeared to reframe the research task itself. Insight generation reached significance in both Fact Synthesis ($p=.019^{*}$, $r=.879$) and Principle Tracking ($p=.043^{*}$, $r=.775$), reflecting a shift from sequential case-by-case filtering toward reasoning about the structure of a legal domain.

P8: *"With Lbox I open case one, skim, decide yes or no. It's factory work. With CaseSense I spent my first two minutes just looking at the graph and could already see data breach cases splitting into external hacking versus internal leaks. That structural insight took two minutes; in the baseline I wouldn't have reached it in twenty."*

P12: *"Instead of reading each case to figure out where it belongs, the graph already showed me the grouping. I could focus on why those groups existed rather than whether they existed."*

6.1.3 Complexity-Dependent Usability Benefits. Usability gains were minimal in Fact Synthesis but pronounced in Principle Tracking, where four of six items reached significance. Mental demand decreased significantly in Principle Tracking ($p=.011^{*}$, $r=-.933$) but not in Fact Synthesis ($p=.127$). Productivity ($p=.022^{*}$, $r=1.000$), satisfaction ($p=.022^{*}$, $r=.838$), and confidence ($p=.037^{*}$, $r=.780$) were higher in Principle Tracking. In Fact Synthesis, only intention to use reached significance ($p=.032^{*}$, $r=.868$), with productivity showing a marginal effect ($p=.062^{\dagger}$, $r=.743$). Intention to use reached marginal significance in Principle Tracking ($p=.051^{\dagger}$, $r=.718$).

P6: *"For housing cooperatives I could have managed without it. For data breaches I genuinely don't think I could have traced the doctrinal evolution in fifteen minutes with keyword tools. The more complex the terrain, the more the map matters."*

One notable finding is that perceived AI reliability showed no significant change in either task (Fact Synthesis: $p=.127$; Principle Tracking: $p=.438$), even as satisfaction and intention to use rose. We attribute this to the two-layer separation: the spatial layer organizes without asserting, so satisfaction could rise without any claim on trust in AI-generated content, while cheap source-sentence verification led participants to verify more rather than trust more.

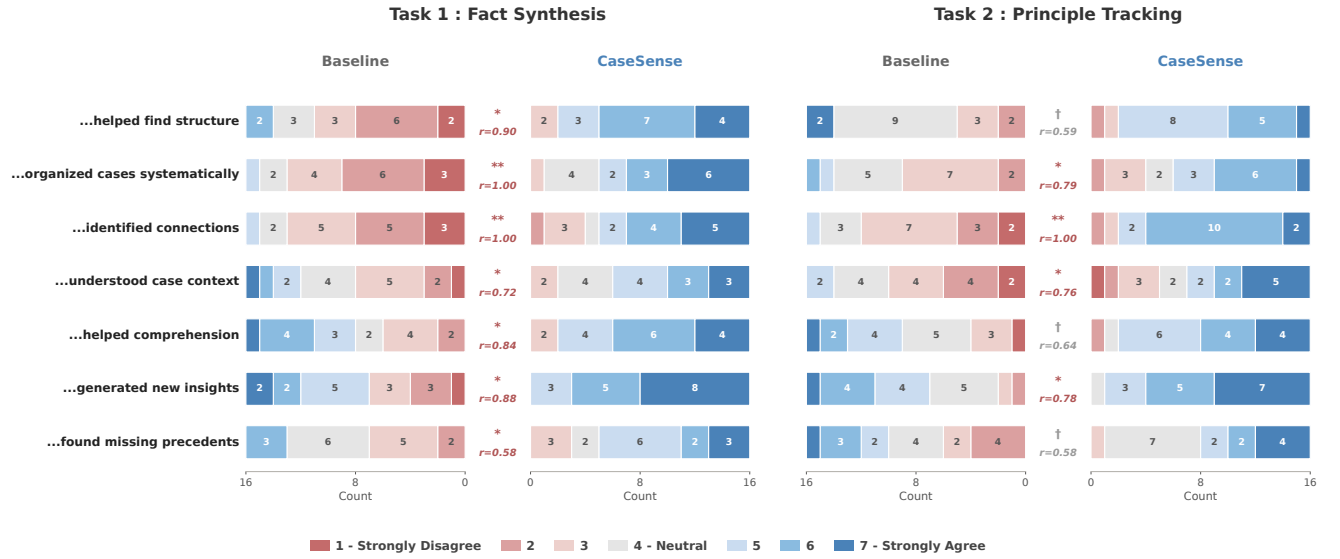


Figure 4: Diverging stacked bar charts comparing sensemaking ratings (7-point Likert) between Baseline and CaseSense across both tasks. Significance markers show Holm–Bonferroni-corrected p -values within the 7-item sensemaking family ($**p < .01$, $*p < .05$, $^{\dagger}p < .10$), with matched-pairs rank-biserial r below each marker.

6.2 RQ2: More Complete & Confident Discovery

CaseSense improved key-case discovery while broadening the scope of exploration (Table 2).

Table 2: Discovery outcomes across conditions and tasks. Searched and Cited: M (SD). Key Disc.: count (rate). Bold-face marks the larger value within each task–metric pair, regardless of condition.

Task	Cond.	Searched	Cited	Key Disc.
Task 1	BL	6.8 (3.2)	3.5 (2.3)	11/16 (68.8%)
	CS	9.1 (4.4)	2.9 (1.7)	16/16 (100%)
Task 2	BL	7.3 (3.0)	4.1 (2.6)	7/16 (43.8%)
	CS	13.0 (7.7)	5.5 (2.5)	14/16 (87.5%)

6.2.1 Reducing Critical Failures. Key-case discovery reached 100% in Fact Synthesis (+31.2 pp over baseline) and 87.5% in Principle Tracking (+43.7 pp). Baseline failures were not random: two participants encountered the key case but failed to recognize its significance because it surfaced late in a sequential list. With CaseSense, 10 of 16 participants discovered the key case within their first three interactions; the retrieval pipeline placed it in the top-5 for 93.8% of sessions.

6.2.2 From Uncertainty to Coverage Awareness. Participants explored more broadly with CaseSense (34% more cases in Fact Synthesis, 78% more in Principle Tracking) and described a shift from anxious searching to confident exploration. P1: “Keyword tools feel like casting a net into dark water. CaseSense makes the water transparent: I can see everything and stop when I’m genuinely done.” P5: “In Front View I immediately noticed a five-year gap with no cases

at all. That absence was itself information; a ranked list never shows you what isn’t there.”

6.2.3 Clustering as Cold-Start Resolution. The cluster overview changed the opening phase of research from disoriented scanning into purposeful navigation. P2: “The real cost of keyword search isn’t searching. It’s the thirty minutes figuring out what the field looks like. Clusters answered that before I clicked a single node.” P16: “When scattered cases are grouped like this, I immediately see the main threads and where they branch.”

6.2.4 Structured Extractions as Efficient Preview. The Case Information Panel’s structured fields served as a rapid triage mechanism, enabling practitioners to assess a case’s relevance without reading the full decision. P14: “I could scan Legal Principle and Facts in five seconds and know whether a case was worth reading in full. With keyword search I’d open the whole decision and skim for three minutes before making the same judgment.”

P13: “The extractions occasionally missed a nuance, but they were reliable enough as a first filter. When something looked promising, I read the original text; the extraction told me where to look.” This triage function helps explain the broader exploration patterns: participants could afford to examine more cases because the cost per case dropped substantially.

6.3 RQ3: Interaction Mechanism Contributions

The three interaction mechanisms contributed not equally but appropriately to each task. Participants spontaneously selected task-relevant dimensions and views without instruction (Table 3; interaction-annotated timelines are in Figure 6, Appendix).

6.3.1 Task-Appropriate Discovery Patterns Without Instruction. Different analytical demands were associated with different techniques,

Table 3: Interaction patterns in the CaseSense condition (M , SD or %). Metrics are descriptive, characterizing how CaseSense was used rather than whether it outperformed baseline. Discovery triggers are multi-label (percentages sum to >100%). The f : p ratio reflects task-driven dimensional emphasis.

Metric	Task 1	Task 2
<i>Interaction Frequency (per session)</i>		
Anchor changes	1.2 (0.4)	2.1 (1.4)
Wt. principle	0.5 (0.8)	1.8 (1.8)
Wt. facts	1.1 (1.1)	0.2 (0.7)
View transitions	1.6 (0.6)	1.9 (0.9)
Cluster clicks	0.6 (0.8)	0.5 (0.9)
<i>Discovery Triggers (% of discoveries; multi-label)</i>		
Anchor change	33.3	40.5
Wt. principle	17.9	35.1
Wt. facts	23.1	5.4
View transition	43.6	45.9
Direct exploration	25.6	18.9
<i>Technique Effectiveness</i>		
f : p ratio	2.2 : 1	0.1 : 1
<i>View Dwell Time (% of session)</i>		
Front (time-princ.)	13.2	51.0
Iso (3D overview)	35.2	44.2
Side (facts-princ.)	51.6	4.8

and participants converged on task-appropriate projections without guidance. View transitions were the most frequent discovery trigger in both tasks (43.6% in Fact Synthesis, 45.9% in Principle Tracking), but the co-occurring techniques diverged sharply. In Fact Synthesis, facts-axis adjustments appeared in 23.1% of discovery windows, more than twice the rate of principle-axis adjustments (17.9%), reflecting the task’s demand for factual comparison. In Principle Tracking, this pattern reversed: principle-axis adjustments appeared in 35.1% of windows versus only 5.4% for facts-axis adjustments, and anchor changes rose to 40.5% (from 33.3% in Fact Synthesis), reflecting the need for perspective shifting when tracing principle evolution. The destination view also differed systematically: Side View dominated Fact Synthesis (51.6% of session time), Front View dominated Principle Tracking (51.0%). This pattern is visually apparent in the interaction-annotated timelines (Appendix, Figure 6): Fact Synthesis sessions are dominated by blue (Side View) backgrounds with facts-axis markers, while Principle Tracking sessions are dominated by rose (Front View) backgrounds with anchor and principle-axis markers.

P10: “Nobody told me to use Front View, but it was obvious. I needed time on one axis and principle on the other. Side View would have been useless; I cared about how the standard evolved, not factual differences.”

P4 showed a distinctive mid-session pattern: an ISO segment followed by a return to the constrained view, a zoom-out-then-refocus strategy that P4 called “stepping back to see the forest before returning to the trees.”

6.3.2 *Weight Sliders as Cognitive Instruments.* The f : p ratio reversed between tasks (2.2 : 1 in Fact Synthesis vs. 0.1 : 1 in Principle Tracking), indicating that practitioners treated weight sliders as intentional instruments for foregrounding the task-relevant axis. P7: “I pushed the slider toward ‘facts’ and three cases that had been scattered snapped into a tight cluster. Same facts, three legal conclusions—exactly the comparison I needed. I assembled that argument in thirty seconds with one slider.”

6.3.3 *Anchoring as Perspective Shifts.* Anchor changes were nearly twice as frequent in Principle Tracking ($M=2.1$) as in Fact Synthesis ($M=1.2$), consistent with the greater need for perspective shifting when tracing principle evolution. P11: “I set the 2017 Supreme Court ruling as anchor and the graph reorganized: cases that seemed unrelated turned out to be close neighbors sharing the same principle foundation but different vocabulary. The anchor didn’t find new cases; it revealed that cases I’d already found were connected.”

6.3.4 *From Linear Search to Sensemaking Loops.* Taken together, these patterns reveal a workflow shift from a linear cycle (search → scan → refine keywords → repeat) into an iterative sensemaking loop (search → visualize → anchor → adjust weights → switch views → discover → refine). P9: “With conventional tools I check each result, adjust keywords, repeat—walking through fog with no sense of progress.” P15: “Keyword research is excavation with a toothbrush—grain by grain, no sense of the whole. CaseSense is ground-penetrating radar: you see the structure first, then decide where to dig.”

6.4 Challenges and Suggestions

Participants identified three usability issues and several directions for improvement.

6.4.1 *Learning Curve and Spatial Literacy.* Three participants (P3, P9, P16) initially struggled with interpreting the ISO View, defaulting to the constrained 2D projections for the first several minutes. P9: “The ISO View felt overwhelming at first. I couldn’t tell which direction meant what until I switched to Front View and the axes made sense.” This suggests that onboarding should foreground the constrained views as entry points before introducing free rotation.

6.4.2 *Node Density and Information Scent.* Two participants failed to discover the key case in Principle Tracking. In both instances, the key case was retrieved (appearing in the top-10) but was positioned in a dense cluster where node overlap obscured it. These failures underscore the node-density issue: retrieval success does not guarantee discovery when visual accessibility is compromised. P10: “Nodes cluster so densely that clicking the right one is hard.” P15: “If nodes showed factual keywords on hover, I’d feel more confident before clicking.” These observations point toward feedforward techniques that would increase information scent without cluttering the visualization.

6.4.3 *Richer Clustering Dimensions.* Participants requested additional ways to organize the case space: factual-element grouping (P4), outcome-based organization (P10), and legal-principle taxonomies (P11). These requests suggest that the current fixed-axis design would benefit from user-configurable axis semantics, allowing practitioners to swap axis dimensions based on their current research question.

7 Discussion

Our findings yield two observations extending beyond the legal domain: how preserving retrieval dimensions as manipulable spatial axes shapes expert reasoning, and how separating spatial organization from AI-generated content preserves verification behavior.

Retrieval Dimensions as Manipulable Axes Turn Visualization into a Reasoning Medium. Most retrieval systems collapse factual similarity, principle alignment, and temporal position into a single relevance score, discarding dimensional structure before users see it. CaseSense preserves these signals as the axes users manipulate, so that the decomposition driving search simultaneously defines the coordinate system practitioners navigate.

Our findings show that participants exploited this manipulability in task-appropriate ways without instruction. In Fact Synthesis, practitioners gravitated toward Side View and adjusted the facts axis far more often than the principle axis; in Principle Tracking, this reversed entirely, with Front View dominating, principle-axis adjustments increasing sharply, and anchor changes nearly doubling. Participants treated weight sliders not as generic controls to explore casually, but as intentional instruments for foregrounding the task-relevant dimension.

When axes encode meanings practitioners already reason with, the visualization becomes a medium for cognitive operations that would otherwise remain in working memory. The spatial layout externalized inter-case relationships and freed attention for evaluation; switching views substituted for mentally re-sorting cases along a different dimension. LLM-augmented sensemaking research has shown that structuring AI outputs into navigable spatial layouts improves exploration [62]; CaseSense extends this by showing that domain-grounded dimensions lead practitioners to converge on task-appropriate strategies without guidance.

This principle generalizes as a transfer hypothesis, with our civil-law scope in mind. Legal research is distinctive in three respects: reasoning decomposes into a stable fact–principle–time triad, precedents can expire so temporal position carries meaning beyond recency, and cross-level fragmentation forces synthesis across documents. Domains sharing these properties—medical literature review (patient presentation, mechanism, publication date, with superseding guidelines) or patent prior-art search (technical features, claim scope, filing date)—are natural targets; the approach presupposes an extractable per-document decomposition and experts accountable for verification, and does not target domains lacking a stable decomposition, such as open-ended web research.

Spatial Structure Supports Without Asserting, Preserving Professional Judgment. A recurring concern with AI-augmented tools in high-stakes domains is that efficiency gains come at the cost of critical judgment: as professionals rely more on AI outputs, they may verify less [6]. Our findings suggest a design strategy that sidesteps this trade-off.

Satisfaction, confidence, and intention to use all increased, yet perceived AI reliability did not change. Participants distinguished between two layers: the spatial layer organizes cases without claiming which are important; the content layer makes factual claims. Participants relied on the layout to guide exploration but routinely verified extracted content against source sentences before citing a

case. By separating the component that organizes from the component that asserts, CaseSense may allow trust in spatial structure to grow without requiring trust in AI-generated text.

This two-layer architecture offers a design principle for expert-facing systems: when AI positions items in a meaningful coordinate system rather than recommending or summarizing, professionals benefit from organizational support while maintaining independent verification. Whether this behavior persists over extended use remains open; future work should investigate whether design interventions such as confidence indicators can sustain critical engagement as practitioners gain fluency.

Limitations and Future Work. Our study has eight limitations, spanning the strength of our evidence and the maturity of the pipeline. The evidence is bounded in scale and scope: $N=16$ yields limited statistical power; 15-minute tasks are shorter than real research sessions, so longitudinal deployment is needed to test whether gains grow with fluency; and the evaluation covered Korean civil law only, leaving other legal traditions open. Measurement adds three caveats: the ecologically valid baseline preserves external validity but limits internal attribution (controlled ablation studies would complement the discovery-trigger analysis); the RQ1 sensemaking construct is self-reported (a graded objective measure against F5’s ground truth would complement the objective RQ2–RQ3 outcomes); and key-case ground truth was defined by a single expert (broader validation would strengthen the metric). Finally, the pipeline is an early instantiation: users cannot inspect or correct query decomposition when the LLM misclassifies a query element, the cluster overlay is single-resolution, and extraction quality was not audited standalone—a decomposition-editing affordance, hierarchical HDBSCAN with an offline interactive cut, and a dedicated extraction audit are future work.

8 Conclusion

CaseSense is a visual analytics system that organizes legal cases along three axes: factual distance, principle distance, and decision date. A within-subjects study ($N=16$) showed improved key-case discovery and sensemaking over conventional workflows, with two findings beyond law. First, participants adopted task-appropriate spatial strategies without instruction: encoding domain reasoning dimensions as manipulable axes makes visualization a tool for thinking, not a display medium. Second, participants valued spatial organization while maintaining skepticism toward AI-generated content: separating structure from content claims sustains professional judgment. These principles apply wherever experts synthesize multi-dimensional relationships under accountability.

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References

- [1] Christopher Ahlberg and Ben Shneiderman. 1994. Visual information seeking: tight coupling of dynamic query filters with starfield displays. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Boston, Massachusetts, USA) (CHI '94). Association for Computing Machinery, New York, NY, USA, 313–317. doi:10.1145/191666.191775
- [2] Kholod Alsufiani, Simon Attfield, and Leishi Zhang. 2017. Towards an Instrument for Measuring Sensemaking and an Assessment of Its Theoretical Features. In *Proceedings of the 31st International BCS Human Computer Interaction Conference (HCI 2017)*. doi:10.14236/ewic/HCI2017.86
- [3] Sophia Althammer, Arian Askari, Suzan Verberne, and Allan Hanbury. 2021. DoSSIER@COLIEE 2021: Leveraging Dense Retrieval and Summarization-Based Re-Ranking for Case Law Retrieval. arXiv:2108.03937 [cs] doi:10.48550/arXiv.2108.03937
- [4] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2023. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. In *The Twelfth International Conference on Learning Representations*.
- [5] Virginia Braun and Victoria Clarke. 2006. Using Thematic Analysis in Psychology. *Qualitative Research in Psychology* 3, 2 (Jan. 2006), 77–101. doi:10.1191/1478088706qp0630a
- [6] Zana Bućina, Maja Barbara Malaya, and Krzysztof Z. Gajos. 2021. To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1, Article 188 (April 2021), 21 pages. doi:10.1145/3449287
- [7] Ricardo J. G. B. Campello, Davoud Moulavi, and Joerg Sander. 2013. Density-Based Clustering Based on Hierarchical Density Estimates. In *Advances in Knowledge Discovery and Data Mining*, Jian Pei, Vincent S. Tseng, Longbing Cao, Hiroshi Motoda, and Guandong Xu (Eds.). Springer, Berlin, Heidelberg, 160–172. doi:10.1007/978-3-642-37456-2_14
- [8] Casenote. [n. d.]. Casenote: Legal Case Information Service. <https://casenote.kr/>.
- [9] Ilias Chalkidis, Manos Fergadiotis, Prodromos Malakasiotis, Nikolaos Aletras, and Ion Androutsopoulos. 2020. LEGAL-BERT: The Muppets Straight out of Law School. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, Trevor Cohn, Yulan He, and Yang Liu (Eds.). Association for Computational Linguistics, Online, 2898–2904. doi:10.18653/v1/2020.findings-emnlp.261
- [10] Ilias Chalkidis, Abhik Jana, Dirk Hartung, Michael Bommarito, Ion Androutsopoulos, Daniel Katz, and Nikolaos Aletras. 2022. LexGLUE: A Benchmark Dataset for Legal Language Understanding in English. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (Eds.). Association for Computational Linguistics, Dublin, Ireland, 4310–4330. doi:10.18653/v1/2022.acl-long.297
- [11] Ran Chen, Xinhuan Shu, Jiahui Chen, Di Weng, Junxiu Tang, Siwei Fu, and Yingcai Wu. 2022. Nebula: A Coordinating Grammar of Graphics. *IEEE Transactions on Visualization and Computer Graphics* 28, 12 (Feb. 2022), 4127–4140. doi:10.1109/TVCG.2021.3076222
- [12] Xi Chen, Wei Zeng, Yanna Lin, Hayder Mahdi Al-manee, Jonathan Roberts, and Remco Chang. 2021. Composition and Configuration Patterns in Multiple-View Visualizations. *IEEE Transactions on Visualization and Computer Graphics* 27, 2 (Feb. 2021), 1514–1524. doi:10.1109/TVCG.2020.3030338
- [13] Matthew Dahl, Varun Magesh, Mirac Suzgun, and Daniel E. Ho. 2024. Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models. *Journal of Legal Analysis* 16, 1 (Jan. 2024), 64–93. arXiv:2401.01301 [cs] doi:10.1093/jla/laae003
- [14] Alex Endert, Patrick Fiaux, and Chris North. 2012. Semantic interaction for visual text analytics. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI '12). Association for Computing Machinery, New York, NY, USA, 473–482. doi:10.1145/2207676.2207741
- [15] Alex Endert, Chao Han, Dipayan Maiti, Leanna House, Scotland Leman, and Chris North. 2011. Observation-Level Interaction with Statistical Models for Visual Analytics. In *2011 IEEE Conference on Visual Analytics Science and Technology (VAST)*. 121–130. doi:10.1109/VAST.2011.6102449
- [16] Rebecca Faust, David Glickenstein, and Carlos Scheidegger. 2019. DimReader: Axis Lines That Explain Non-Linear Projections. *IEEE Transactions on Visualization and Computer Graphics* 25, 1 (Jan. 2019), 481–490. doi:10.1109/TVCG.2018.2865194
- [17] Yi Feng, Chuanyi Li, and Vincent Ng. 2024. Legal Case Retrieval: A Survey of the State of the Art. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 6472–6485. doi:10.18653/v1/2024.acl-long.350
- [18] Daniel Fürst, Mennatallah El-Assady, Daniel A. Keim, and Maximilian T. Fischer. 2025. Challenges and Opportunities for Visual Analytics in Jurisprudence. *Artificial Intelligence and Law* (Nov. 2025). doi:10.1007/s10506-025-09494-2
- [19] Katy Ilonka Gero, Chelse Swoopes, Ziwei Gu, Jonathan K. Kummerfeld, and Elena L. Glassman. 2024. Supporting Sensemaking of Large Language Model Outputs at Scale. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–21. doi:10.1145/3613904.3642139
- [20] Google. 2025. LangExtract: A Python library for extracting structured information from unstructured text using LLMs. <https://github.com/google/langextract>. Accessed: 2026-03-28.
- [21] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research. In *Advances in Psychology*, Peter A. Hancock and Najmedin Meshkati (Eds.). Human Mental Workload, Vol. 52. North-Holland, 139–183. doi:10.1016/S0166-4115(08)62386-9
- [22] Wonseok Hwang, Dongjun Lee, Kyoungyeon Cho, Hanuul Lee, and Minjoon Seo. 2022. A Multi-Task Benchmark for Korean Legal Language Understanding and Judgement Prediction. *Advances in Neural Information Processing Systems* 35 (Dec. 2022), 32537–32551.
- [23] Paulo Joia, Danilo Coimbra, Jose A. Cuminato, Fernando V. Paulovich, and Luis G. Nonato. 2011. Local Affine Multidimensional Projection. *IEEE Transactions on Visualization and Computer Graphics* 17, 12 (Feb. 2011), 2563–2571. doi:10.1109/TVCG.2011.220
- [24] Hyeonsu B Kang, Tongshuang Wu, Joseph Chee Chang, and Aniket Kittur. 2023. Synergi: A Mixed-Initiative System for Scholarly Synthesis and Sensemaking. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) (UIST '23). Association for Computing Machinery, New York, NY, USA, Article 43, 19 pages. doi:10.1145/3586183.3606759
- [25] Dave S Kerby. 2014. The simple difference formula: An approach to teaching nonparametric correlation. *Comprehensive Psychology* 3 (2014), 11–17.
- [26] Chaeun Kim, Jinu Lee, and Wonseok Hwang. 2025. LegalSearchLM: Rethinking Legal Case Retrieval as Legal Elements Generation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (Eds.). Association for Computational Linguistics, Suzhou, China, 4521–4554. doi:10.18653/v1/2025.emnlp-main.225
- [27] Hannah Kim, Jaegul Choo, Haesun Park, and Alex Endert. 2016. InterAxis: Steering Scatterplot Axes via Observation-Level Interaction. *IEEE transactions on visualization and computer graphics* 22, 1 (Jan. 2016), 131–140. doi:10.1109/TVCG.2015.2467615
- [28] Junghwan Kim, Hyeonseok Jeon, Dongseok Heo, Jung Lee, and Bongwon Suh. 2025. LegisFlow: Enhancing Korean Legal Research with Temporal-Aware LLM Interfaces. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology* (UIST '25). Association for Computing Machinery, New York, NY, USA, Article 142, 29 pages. doi:10.1145/3746059.3747752
- [29] Yeeun Kim, Youngrok Choi, Eunkyung Choi, JinHwan Choi, Hai Jin Park, and Wonseok Hwang. 2024. Developing a Pragmatic Benchmark for Assessing Korean Legal Language Understanding in Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, Yaser Al-Otaiz, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 5573–5595. doi:10.18653/v1/2024.findings-emnlp.319
- [30] G. Klein, B. Moon, and R.R. Hoffman. 2006. Making Sense of Sensemaking 1: Alternative Perspectives. *IEEE Intelligent Systems* 21, 4 (July 2006), 70–73. doi:10.1109/MIS.2006.75
- [31] C.C. Kuhlthau and S.L. Tama. 2001. Information Search Process of Lawyers: A Call for 'just for Me' Information Services. *Journal of Documentation* 57, 1 (Feb. 2001), 25–43. doi:10.1108/EUM00000000007076
- [32] Bum Chul Kwon, Hannah Kim, Emily Wall, Jaegul Choo, Haesun Park, and Alex Endert. 2017. AxiSketcher: Interactive Nonlinear Axis Mapping of Visualizations through User Drawings. *IEEE Transactions on Visualization and Computer Graphics* 23, 1 (2017), 221–230. doi:10.1109/TVCG.2016.2598446
- [33] Michelle S. Lam, Janice Teoh, James A. Landay, Jeffrey Heer, and Michael S. Bernstein. 2024. Concept Induction: Analyzing Unstructured Text with High-Level Concepts Using LLoM. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 766, 28 pages. doi:10.1145/3613904.3642830
- [34] Law & Business. [n. d.]. LawNBNB: Legal Information Service. <https://www.lawnb.com/>.
- [35] Lbox. [n. d.]. Lbox: Legal Research Platform. <https://lbox.kr/v2>.
- [36] Sam Yu-Te Lee and Kwan-Liu Ma. 2025. HINTs: Sensemaking on Large Collections of Documents with Hypergraph Visualization and Intelligent Agents. *IEEE Transactions on Visualization and Computer Graphics* 31, 9 (Sept. 2025), 5532–5546. arXiv:2403.02752 [cs] doi:10.1109/TVCG.2024.3459961
- [37] James R. Lewis. 1992. Psychometric Evaluation of the Post-Study System Usability Questionnaire: The PSSUQ. *Proceedings of the Human Factors Society Annual Meeting* 36, 16 (Oct. 1992), 1259–1260. doi:10.1177/154193129203601617
- [38] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems*, Vol. 33. Curran Associates, Inc., 9459–9474.
- [39] Haitao Li, Qingyao Ai, Jia Chen, Qian Dong, Yueyue Wu, Yiqun Liu, Chong Chen, and Qi Tian. 2023. SAILER: Structure-aware Pre-trained Language Model for Legal Case Retrieval. In *Proceedings of the 46th International ACM SIGIR Conference*

- on Research and Development in Information Retrieval (Taipei, Taiwan) (SIGIR '23). Association for Computing Machinery, New York, NY, USA, 1035–1044. doi:10.1145/3539618.3591761
- [40] Matte Lim, Catherine Yeh, Martin Wattenberg, Fernanda Viégas, and Panagiotis Michalatos. 2025. Chronotome: Real-Time Topic Modeling for Streaming Embedding Spaces. 181–185 pages.
- [41] Michael Xieyang Liu, Tongshuang Wu, Tianying Chen, Franklin Mingzhe Li, Aniket Kittur, and Brad A Myers. 2024. Selenite: Scaffolding Online Sensemaking with Comprehensive Overviews Elicited from Large Language Models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 837, 26 pages. doi:10.1145/3613904.3642149
- [42] Yang Liu, Eunice Jun, Qisheng Li, and Jeffrey Heer. 2019. Latent Space Cartography: Visual Analysis of Vector Space Embeddings. *Computer Graphics Forum* 38 (July 2019), 67–78. doi:10.1111/cgf.13672
- [43] Chuan Luo, Xin (Robert) Luo, Laurie Schatzberg, and Choon Ling Sia. 2013. Impact of Informational Factors on Online Recommendation Credibility: The Moderating Role of Source Credibility. *Decision Support Systems* 56 (Dec. 2013), 92–102. doi:10.1016/j.dss.2013.05.005
- [44] Varun Magesh, Faiz Surani, Matthew Dahl, Mirac Suzgun, Christopher D. Manning, and Daniel E. Ho. 2025. Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools. *Journal of Empirical Legal Studies* 22, 2 (2025), 216–242. doi:10.1111/jels.12413
- [45] Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. *Introduction to Information Retrieval*. Cambridge university press, Cambridge.
- [46] Leland McInnes, John Healy, and James Melville. 2020. UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. arXiv:1802.03426 [stat] doi:10.48550/arXiv.1802.03426
- [47] Open Law. [n. d.]. Open Law: Legal Information System. <https://open.law.go.kr/LSO/main.do>.
- [48] Jeongeun Park, Bryan Min, Kihoon Son, Jean Y. Song, Xiaojuan Ma, and Juho Kim. 2025. ChoiceMates: Supporting Unfamiliar Online Decision-Making with Multi-Agent Conversational Interactions. arXiv:2310.01331 [cs] doi:10.48550/arXiv.2310.01331
- [49] Nicholas Pipitone and Ghita Hour Alami. 2024. LegalBench-RAG: A Benchmark for Retrieval-Augmented Generation in the Legal Domain. arXiv:2408.10343 [cs] doi:10.48550/arXiv.2408.10343
- [50] Peter Pirolli and Stuart Card. 2005. The sensemaking process and leverage points for analyst technology as identified through cognitive task analysis. 5, 1 (2005), 2–4.
- [51] Lucas E. Resck, Jean R. Ponciano, Luis Gustavo Nonato, and Jorge Poco. 2023. LegalVis: Exploring and Inferring Precedent Citations in Legal Documents. *IEEE Transactions on Visualization and Computer Graphics* 29, 6 (June 2023), 3105–3120. doi:10.1109/TVCG.2022.3152450
- [52] Stephen Robertson and Hugo Zaragoza. 2009. The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval* 3, 4 (2009), 333–389. doi:10.1561/15000000019
- [53] Daniel M. Russell, Mark J. Stefik, Peter Pirolli, and Stuart K. Card. 1993. The cost structure of sensemaking. In *Proceedings of the INTERACT '93 and CHI '93 Conference on Human Factors in Computing Systems* (Amsterdam, The Netherlands) (CHI '93). Association for Computing Machinery, New York, NY, USA, 269–276. doi:10.1145/169059.169209
- [54] Cheol Ryu, Seolhwa Lee, Subeen Pang, Chanyeol Choi, Hojun Choi, Myeonggee Min, and Jy-Yong Sohn. 2023. Retrieval-Based Evaluation for LLMs: A Case Study in Korean Legal QA. In *Proceedings of the Natural Legal Language Processing Workshop 2023*, Daniel Preotiu-Pietro, Catalina Goanta, Ilias Chalkidis, Leslie Barrett, Gerasimos Spanakis, and Nikolaos Aletras (Eds.). Association for Computational Linguistics, Singapore, 132–137. doi:10.18653/v1/2023.nllp-1.13
- [55] Jessica Zeitz Self, Michelle Dowling, John Wenskovich, Ian Crandell, Ming Wang, Leanna House, Scotland Leman, and Chris North. 2018. Observation-Level and Parametric Interaction for High-Dimensional Data Analysis. *ACM Trans. Interact. Intell. Syst.* 8, 2, Article 15 (June 2018), 36 pages. doi:10.1145/3158230
- [56] Kiho Seo and Takehito Utsuro. 2025. RAG Based Question Answering of Korean Laws and Precedents. In *Proceedings of the Eighth Fact Extraction and VERification Workshop (FEVER)*, Mubashara Akhtar, Rami Aly, Christos Christodoulopoulos, Oana Cocarascu, Zhijiang Guo, Arpit Mittal, Michael Schlichtkrull, James Thorne, and Andreas Vlachos (Eds.). Association for Computational Linguistics, Vienna, Austria, 91–100. doi:10.18653/v1/2025.fever-1.7
- [57] Yunqiu Shao, Jiaxin Mao, Yiqun Liu, Weizhi Ma, Ken Satoh, Min Zhang, and Shaoping Ma. 2020. BERT-PLI: Modeling Paragraph-Level Interactions for Legal Case Retrieval. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence*. International Joint Conferences on Artificial Intelligence Organization, Yokohama, Japan, 3501–3507. doi:10.24963/ijcai.2020/484
- [58] B. Shneiderman. 1994. Dynamic Queries for Visual Information Seeking. *IEEE Software* 11, 6 (Jan. 1994), 70–77. doi:10.1109/52.329404
- [59] Eero Sormunen, Norbert Erdmann, Suzanne Csa Otieno, Mirjamajja Mikkilä-Erdmann, Eero Laakkonen, Teemu Mikkonen, Md Arman Hossain, Roberto González-Ibáñez, Mario Quintanilla-Gatica, Paavo Ht Leppänen, and Marja Vauras. 2023. How Do Gender, Internet Activity and Learning Beliefs Predict Sixth-Grade Students' Self-Efficacy Beliefs in and Attitudes towards Online Inquiry? *Journal of Information Science* 49, 5 (Oct. 2023), 1246–1261. doi:10.1177/01655515211043708
- [60] Julian Stahnke, Marian Dörk, Boris Müller, and Andreas Thom. 2016. Probing Projections: Interaction Techniques for Interpreting Arrangements and Errors of Dimensionality Reductions. *IEEE Transactions on Visualization and Computer Graphics* 22, 1 (Jan. 2016), 629–638. doi:10.1109/TVCG.2015.2467717
- [61] Sangho Suh, Meng Chen, Bryan Min, Toby Jia-Jun Li, and Haijun Xia. 2024. Luminate: Structured Generation and Exploration of Design Space with Large Language Models for Human-AI Co-Creation. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 644, 26 pages. doi:10.1145/3613904.3642400
- [62] Sangho Suh, Bryan Min, Srishti Palani, and Haijun Xia. 2023. Sensecape: Enabling Multilevel Exploration and Sensemaking with Large Language Models. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) (UIST '23). Association for Computing Machinery, New York, NY, USA, Article 1, 18 pages. doi:10.1145/3586183.3606756
- [63] Yanran Tang, Ruihong Qiu, Yilun Liu, Xue Li, and Zi Huang. 2024. CaseGNN++: Graph Contrastive Learning for Legal Case Retrieval with Graph Augmentation. arXiv:2405.11791 [cs] doi:10.48550/arXiv.2405.11791
- [64] Yanran Tang, Ruihong Qiu, Hongzhi Yin, Xue Li, and Zi Huang. 2024. CaseLink: Inductive Graph Learning for Legal Case Retrieval. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Washington DC, USA) (SIGIR '24). Association for Computing Machinery, New York, NY, USA, 2199–2209. doi:10.1145/3626772.3657693
- [65] Chun-Hua Tsai and Peter Brusilovsky. 2021. The Effects of Controllability and Explainability in a Social Recommender System. *User Modeling and User-Adapted Interaction* 31, 3 (July 2021), 591–627. doi:10.1007/s11257-020-09281-5
- [66] Laurens van der Maaten, Geoffrey Hinton, and Yoesoep Rachmad. 2008. Visualizing Data Using T-SNE. *Journal of Machine Learning Research* 9 (Nov. 2008), 2579–2605.
- [67] Qianwen Wang, Sehi L'Yi, and Nils Gehlenborg. 2023. DRAVA: Aligning Human Concepts with Machine Learning Latent Dimensions for the Visual Exploration of Small Multiples. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, Hamburg Germany, 1–15. doi:10.1145/3544548.3581127
- [68] Nirmalie Wiratunga, Ramitha Abeyratne, Lasal Jayawardena, Kyle Martin, Stewart Massie, Ikechukwu Nkisi-Orji, Ruwan Weerasinghe, Anne Liret, and Bruno Fleisch. 2024. CBR-RAG: Case-Based Reasoning for Retrieval Augmented Generation in LLMs for Legal Question Answering. In *Case-Based Reasoning Research and Development*, Juan A. Recio-Garcia, Mauricio G. Orozco-del-Castillo, and Derek Bridge (Eds.). Springer Nature Switzerland, Cham, 445–460. doi:10.1007/978-3-031-63646-2_29
- [69] Wayne Xin Zhao, Jing Liu, Ruiyang Ren, and Ji-Rong Wen. 2024. Dense Text Retrieval Based on Pretrained Language Models: A Survey. *ACM Trans. Inf. Syst.* 42, 4, Article 89 (Feb. 2024), 60 pages. doi:10.1145/3637870
- [70] Yuheng Zhao, Yixing Zhang, Yu Zhang, Xinyi Zhao, Junjie Wang, Zekai Shao, Catayay Turkey, and Siming Chen. 2025. LEVA: Using Large Language Models to Enhance Visual Analytics. *IEEE Transactions on Visualization and Computer Graphics* 31, 3 (March 2025), 1830–1847. arXiv:2403.05816 [cs] doi:10.1109/TVCG.2024.3368060

A Participant Demographic

We recruited 16 licensed Korean attorneys (10 female, 6 male; aged 27–33, $M=30.3$, $SD=2.3$) who had passed the Korean Bar Examination and were actively practicing in civil law. Experience ranged from 1 to 8 years ($M=3.9$, $SD=1.8$), covering career stages from junior associates still building retrieval habits to senior practitioners with established sensemaking strategies. All participants used professional case databases (CaseNote, Lbox, or Lawnb) in their practice; Table 4 reports the tools each participant chose in the Baseline condition, where two relied primarily on web search, so the Baseline reflects each practitioner’s genuine workflow rather than a prescribed toolset. Table 4 summarizes individual demographics.

Table 4: Participant demographics, including the case databases and AI tools each participant reported using in the Baseline condition.

P_ID	Gender	Age	Experience (yrs)	Case DB	AI Tools
P01	Female	33	4	Lbox, CaseNote	Perplexity
P02	Male	31	5	Lbox, CaseNote, Lawnb	None
P03	Female	30	5	Lbox, CaseNote	ChatGPT, Gemini
P04	Female	33	8	Lbox, CaseNote	ChatGPT, SuperLawyer
P05	Female	28	4	— ^a	Perplexity
P06	Male	32	7	CaseNote, Lbox	Lbox AI, ChatGPT
P07	Female	30	4	Lbox, Lawnb	Lbox AI, Gemini
P08	Female	27	3	Lbox, CaseNote	Lbox AI
P09	Female	27	1	Lbox	None
P10	Male	32	4	CaseNote	ChatGPT
P11	Female	27	2	Lbox, CaseNote	Lbox AI, ChatGPT, Gemini
P12	Male	33	5	Lbox, CaseNote	Gemini
P13	Male	29	2	Lbox	ChatGPT, Perplexity
P14	Male	31	2	CaseNote	ChatGPT
P15	Female	33	4	Lawnb, CaseNote	None
P16	Female	28	2	— ^a	ChatGPT

“None” indicates no regular AI tool use. ^a Relied primarily on web search rather than a dedicated case database during the Baseline session.

B Prompt Design

This appendix presents the prompts comprising the CaseSense backend pipeline: structured field extraction (LangExtract), thematic cluster labeling, and query decomposition. Placeholders are shown in {blue}. All prompts are written in English; Korean-language content in outputs is transliterated for readability.

B.1 LangExtract: Structured Field Extraction

LangExtract decomposes a raw Korean legal case into five semantically distinct fields, covering the full logical arc of a decision: issue → principle → fact → reasoning → decision. All extracted spans must appear verbatim in the source document; paraphrase and reconstruction are explicitly prohibited. Related entries across fields are linked via a shared `issue_group` identifier.

Prompt: Structured Field Extraction

System

You are a Korean Legal Case Extraction Specialist responsible for decomposing raw Korean legal case documents into five semantically distinct fields. Your task is to identify and extract verbatim spans from the source document that correspond to each extraction category.

****Fields to extract:****

- ****legal_issue**** (contested point): The central contested legal question presented in the court's holdings section. Phrased as a question typically ending in "whether [X] is the case (affirmative / negative)." Attributes: `issue_id` (identifier); `court_answer` (affirmative / negative).
- ****legal_principle**** (legal principle): The legal principle or interpretive reasoning presented in the judgment summary and reasoning section. Identified by sentences reflecting the court's interpretive stance, e.g., "it shall be deemed that ~," "it must be viewed that ~." Attributes: `issue_group`; `reasoning_type` (interpretive method or basis).
- ****fact**** (factual circumstances): The concrete factual circumstances as determined by the court. Identified by objective factual descriptions typically introduced by "according to the facts established by the court below" or "according to the record." Attributes: `issue_group`; `actor` (relevant party).
- ****court_reasoning**** (court's judgment): The court's specific reasoning process applying legal principles to established facts. Identified by argumentative passages following connectives such as "if so" or "therefore." Attributes: `issue_group`; `reasoning_type` (e.g., factual application, legal subsumption).
- ****court_decision**** (final ruling): The final ruling as stated in the disposition, including the outcome (e.g., appeal dismissed / granted) and allocation of litigation costs. Attributes: `result` (ruling outcome); `winning_party` (prevailing party).

****Constraints:****

- All extracted text must appear verbatim in the source - do ****not**** paraphrase, summarize, or reconstruct.
- Link related entries across all five fields via a consistent `issue_group` identifier.
- Preserve all original legal terminology without translation or substitution.
- Do not infer or reconstruct text not explicitly present in the source document.

****Output format:**** A structured JSON object containing all five extraction categories, each as an array of entries with `extraction_text` and field-specific attributes.

User

{raw_case_text}

Assistant

(Structured JSON with five extraction fields; each entry contains `extraction_text` and field-specific attributes.)

Example output (abbreviated; one issue group shown)

```
{
  "Seoul_High_Court_2010Na77530": {
    "extractions": [
      {
        "extraction_class": "legal_issue",
        "extraction_text": "Whether the delayed response by the defendant
          internet service company to the plaintiff's account theft report
          constitutes an attributable cause for the plaintiff's item loss.",
        "attributes": {
          "issue_group": "ISSUE_01",
          "court_answer": "negative"
        }
      },
      {
        "extraction_class": "fact",
        "extraction_text": "On Sunday, November 1, 2009, at approximately
          17:34, the plaintiff's game account was hijacked, resulting in a
          forced logout. The plaintiff regained access by resetting the
          password via mobile authentication at 17:38, only to find that
          all items had already been removed from the account.",
        "attributes": {
          "issue_group": "ISSUE_01",
          "actor": "plaintiff, defendant"
        }
      },
      {
        "extraction_class": "legal_principle",
        "extraction_text": "The account holder bears sole responsibility
          for managing their account and password. Any damages arising from
          negligent management or unauthorized third-party use are
          attributable to the account holder, not the company.",
        "attributes": {
          "issue_group": "ISSUE_01",
          "reasoning_type": "self-responsibility principle under terms of service"
        }
      },
      {
        "extraction_class": "court_reasoning",
        "extraction_text": "The 24-hour service obligation pertains to
          maintaining the central server and defending against external
          intrusions, but does not extend to requiring staff to be on
          standby around the clock to handle account theft reports arising
          from users' own negligence.",
        "attributes": {
          "issue_group": "ISSUE_01",
          "reasoning_type": "limitation of duty of care based on service characteristics"
        }
      },
      {
        "extraction_class": "court_decision",
        "extraction_text": "The portion of the first instance judgment
          against the defendant is reversed, and the corresponding claims
          by the plaintiff are dismissed. All litigation costs are borne
          by the plaintiff.",
        "attributes": {
          "result": "reversal of first instance and dismissal of claims",
          "winning_party": "defendant"
        }
      }
    ]
  }
}
```

B.2 Thematic Cluster Labeling

After HDBSCAN clusters are formed from *legal_issue*, this prompt generates a concise noun-phrase label for each cluster. Labels serve as visualization node group annotations and must be discriminative across adjacent clusters.

Prompt: Cluster Labeling

System

You are a Korean Legal Taxonomy Specialist responsible for generating concise thematic labels for clusters of Korean legal cases. Your task is to analyze a set of *legal_issue* texts belonging to the same cluster and produce a single noun phrase that captures the shared legal theme.

****Output fields:****

- ****cluster_label****: A concise English noun phrase (3-6 words) in title case summarizing the shared legal theme across all provided texts. Should reflect the common factual or principle pattern, not individual case specifics. Must be discriminative enough to distinguish this cluster from adjacent ones.
- ****cluster_rationale****: A single sentence explaining why the provided issues share this label, grounded in recurring terminology or legal patterns observed across the inputs.

****Constraints:****

- *cluster_label* must be a noun phrase - not a full sentence or question.
- Do not reproduce individual case identifiers or case-specific proper nouns in the label.
- Prioritize factual pattern over statutory citation when naming clusters.
- Maintain consistency in granularity across labels produced in the same batch.
- If the provided issues are too heterogeneous to form a coherent theme, return "Other" with a rationale explaining the heterogeneity.

****Output format:**** {"cluster_label": "...", "cluster_rationale": "..."}

User

{legal_issue_texts}

Assistant

(JSON with cluster_label and cluster_rationale.)

Example

Input:

```
{
  "issues": [
    "Whether a personal data controller is liable for damages when user personal information is leaked due to external attacks such as hacking",
    "Whether negligence of an information and communications service provider is recognized when personal data is leaked through hacking of its database by an external intruder"
  ]
}
```

Output:

```
{
  "cluster_label": "External Data Breach via Hacking",
  "cluster_rationale": "All issues concern liability for personal information leakage resulting from unauthorized external intrusion into the defendant's systems."
}
```

B.3 Query Decomposition

The language model decomposes the natural language query into a *facts_query* capturing factual circumstances and a *principle_query* identifying the relevant legal principle. This separation is architecturally necessary: *facts_query* drives the first-pass ANN retrieval against facts_db, while *principle_query* is used to compute principle similarity $s_p(c)$ against candidates in the second pass—two distinct operations that require the components to remain separate.

Prompt: Query Decomposition

System

You are a Korean Legal Query Analyst responsible for decomposing a natural language search query into two semantically distinct retrieval components. Your task is to separate the factual circumstances embedded in the query from the legal principle it implicates, enabling independent vector retrieval against separate index collections.

Output fields:

- **facts_query**: A concise description of the underlying factual circumstances implied by the query - who did what to whom, under what conditions, and with what outcome. Written in the style of a fact extraction: objective, event-focused, and party-specific. Do **not** include legal conclusions, statutory references, or principle terms.
- **principle_query**: A concise description of the legal principle, rule, or standard implicated by the query - the applicable duty, right, liability standard, or interpretive question at stake. Written in the style of a legal principle extraction: normative, principle-focused, and generalized beyond specific parties. Do **not** include case-specific facts or named parties.

Constraints:

- The two fields must be semantically non-overlapping: factual content belongs exclusively in facts_query; principle content belongs exclusively in principle_query.
- Each field must be 1-3 sentences; do not over-specify.
- When a specific statute or legal standard is named in the query, place it in principle_query even if the user phrased it as a factual condition.

Output format: {"facts_query": "...", "principle_query": "..."}

User

{user_query}

Assistant

(JSON with facts_query, principle_query.)

Example

Input:

```
{
  "query": "Can an online game company be held liable for failing to respond promptly to an account theft report submitted by a user on a Sunday?"
}
```

Output:

```
{
  "facts_query": "A user's game account was hijacked by a third party. The user filed a theft report on a Sunday; the company did not respond until the following business day, by which time in-game items were lost.",
  "principle_query": "Whether a service provider's duty of care requires 24-hour response to account theft reports, including weekends."
}
```

C Technical Evaluation Details

This appendix documents the evaluation protocol and representative tasks used to validate CaseSense’s retrieval and clustering pipeline prior to the user study. Evaluation covered two components: retrieval quality, assessed automatically against expert-established ground truth, and cluster coherence, assessed through independent expert rating.

C.1 Task Construction

We constructed 40 legal research tasks spanning four civil law domains: contract disputes, tort liability, property rights, and data protection (10 tasks per domain). Tasks were designed at three complexity levels—simple (single legal issue, clear factual pattern), moderate (multiple issues or ambiguous facts), and complex (overlapping principles or sparse case)—with each domain contributing tasks at all three levels. Each task was formulated as a natural language query reflecting how practitioners typically frame legal questions. F5 (judicial researcher, 5 years) independently identified the expected key case (anchor ground truth) and a reference set of relevant cases (retrieval ground truth) for each task, drawing on professional legal databases and domain expertise. Table 5 presents one representative task per domain.

Table 5: Representative evaluation tasks by domain.

Domain	Query	Ground Truth	
Contract disputes	“Cases where a contractor was held liable for property damage caused by subcontractor negligence despite the absence of a direct contractual relationship”	Supreme Court 2015Da-XXXXXX	
Tort liability	“Cases where a company was held liable for damages caused by an employee’s personal data breach despite the breach occurring outside working hours”	Supreme Court 2016Da-XXXXXX	
Property rights	“Cases where a tenant was denied compensation for improvements made to a property following wrongful eviction by the landlord”	Seoul High Court 2018Na-XXXXXX	
Data protection	“Cases where failure to implement internal access controls led to corporate liability for customer data leakage by a contracted third party”	Supreme Court 2018Da-XXXXXX	

C.2 Retrieval Evaluation

Retrieval quality was assessed automatically by comparing system output against F5’s ground truth. Three metrics were computed: Precision@5 (proportion of relevant cases in the top-5 candidates), Anchor Accuracy@1 (whether the top-ranked candidate matched the expected key case), and Mean Reciprocal Rank (MRR, position of the first relevant case in the candidate set). The five anchor misses all occurred in complex tasks where factually similar cases applied substantially different legal principles, causing the combined score $\alpha \cdot s_f + \beta \cdot s_p$ to favor a principle-aligned but factually weaker candidate over the expected key case.

C.3 Cluster Coherence Evaluation

F4 (associate, 3 years) and F5 independently evaluated each generated cluster on two criteria: (1) whether cases within the cluster shared legally meaningful relationships, rated as “coherent” or “incoherent”; and (2) whether the generated cluster label faithfully represented the cluster’s legal theme, rated as accurate or inaccurate. Evaluators resolved disagreements through discussion. Inter-rater agreement prior to discussion was substantial across both criteria (Fleiss’ $\kappa=0.76$). The 15.0% of clusters rated incoherent predominantly arose from the HDBSCAN noise category (“Other”), which by design aggregates low-frequency cases without a dominant shared theme. Label inaccuracies (20.0%) were concentrated in clusters where the dominant legal issue obscured secondary factual distinctions—a limitation evaluators identified as a direction for future work.

D Evaluation Instruments

D.1 Post-Study Questionnaire Items

Table 6 lists the 13 post-task items used after each condition–task pair. Items Q1–Q6 assess usability and workload; items Q7–Q13 assess sensemaking capabilities. All items used 7-point Likert scales. Q1 (mental demand) is reverse-scored so that higher values indicate lower demand, aligning directionality with the remaining items.

Table 6: Post-Study Questionnaire Items for Usability, Workload, and Sensemaking

ID	Questionnaire Item (7-point Likert Scale: 1=Strongly Disagree, 7=Strongly Agree, unless otherwise noted)	Ref
Q1	The mental demand required to collaborate with the system was high. (Reverse-scored for analysis) (1=Very little effort needed, 7=Very much effort needed)	[21]
Q2	I believe using this system would make me productive.	[37]
Q3	Overall, I am satisfied with this system.	[37]
Q4	The information provided by this system was reliable.	[43]
Q5	I felt confident finding the necessary information using this system.	[59]
Q6	I intend to use this system in the future when exploring case law information.	[65]
Q7	It helped me find structure in the information.	[2]
Q8	I was able to discover ways to classify/group/annotate cases systematically.	[2]
Q9	I could identify/organize connections between the current issue and past cases.	[2]
Q10	I could understand connections between case backgrounds such as instances, courts, and jurisdictions.	[2]
Q11	This system was practically helpful in understanding the given information.	[2]
Q12	I gained new insights through this system.	[2]
Q13	I could discover missing or insufficient grounds/cases (gaps) in my current analysis.	[2]

D.2 Usability and Workload Results

Figure 5 compares usability and workload ratings across conditions. Usability gains concentrated in Principle Tracking, the more complex task: mental demand decreased significantly ($p < .05$, $r = -.933$), and productivity, satisfaction, and confidence all reached significance. In Fact Synthesis, only intention to use was significant, with productivity at a marginal level. Perceived AI reliability showed no significant change in either task, even as satisfaction and intention to use rose—consistent with the interpretation that CaseSense’s spatial scaffolding supports organization without requiring increased trust in AI-generated content (§6.1).

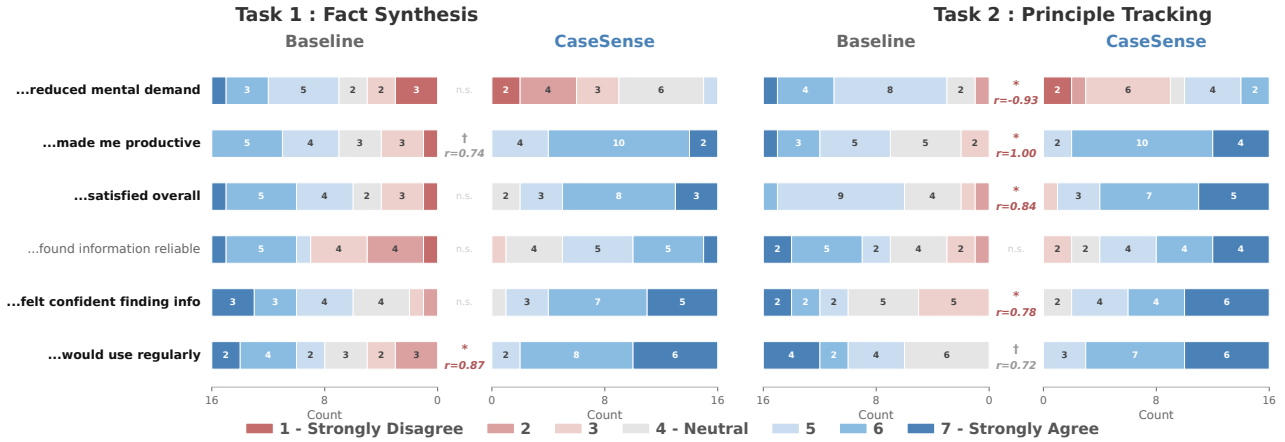


Figure 5: Diverging stacked bar charts comparing usability and workload ratings (7-point Likert) between Baseline and CaseSense across both tasks. Mental demand is reverse-scored (lower = better). Significance markers show Holm–Bonferroni-corrected p -values within the 6-item usability family (* $p < .05$, † $p < .10$), with matched-pairs rank-biserial r below each marker.

D.3 Interaction-Annotated Discovery Timelines

Figure 6 presents session-level interaction timelines for all 16 participants across both tasks. Background color encodes the active view (rose = Front, slate = ISO, blue = Side); markers above each bar indicate anchor changes and weight adjustments; markers on the bar denote the case events defined in §5 (searched, cited, key case). The visual pattern corroborates the task-driven projection selection reported in §6.3: Fact Synthesis sessions are dominated by blue (Side View) backgrounds with facts-axis markers, while Principle Tracking sessions are dominated by rose (Front View) backgrounds with anchor and principle-axis markers.

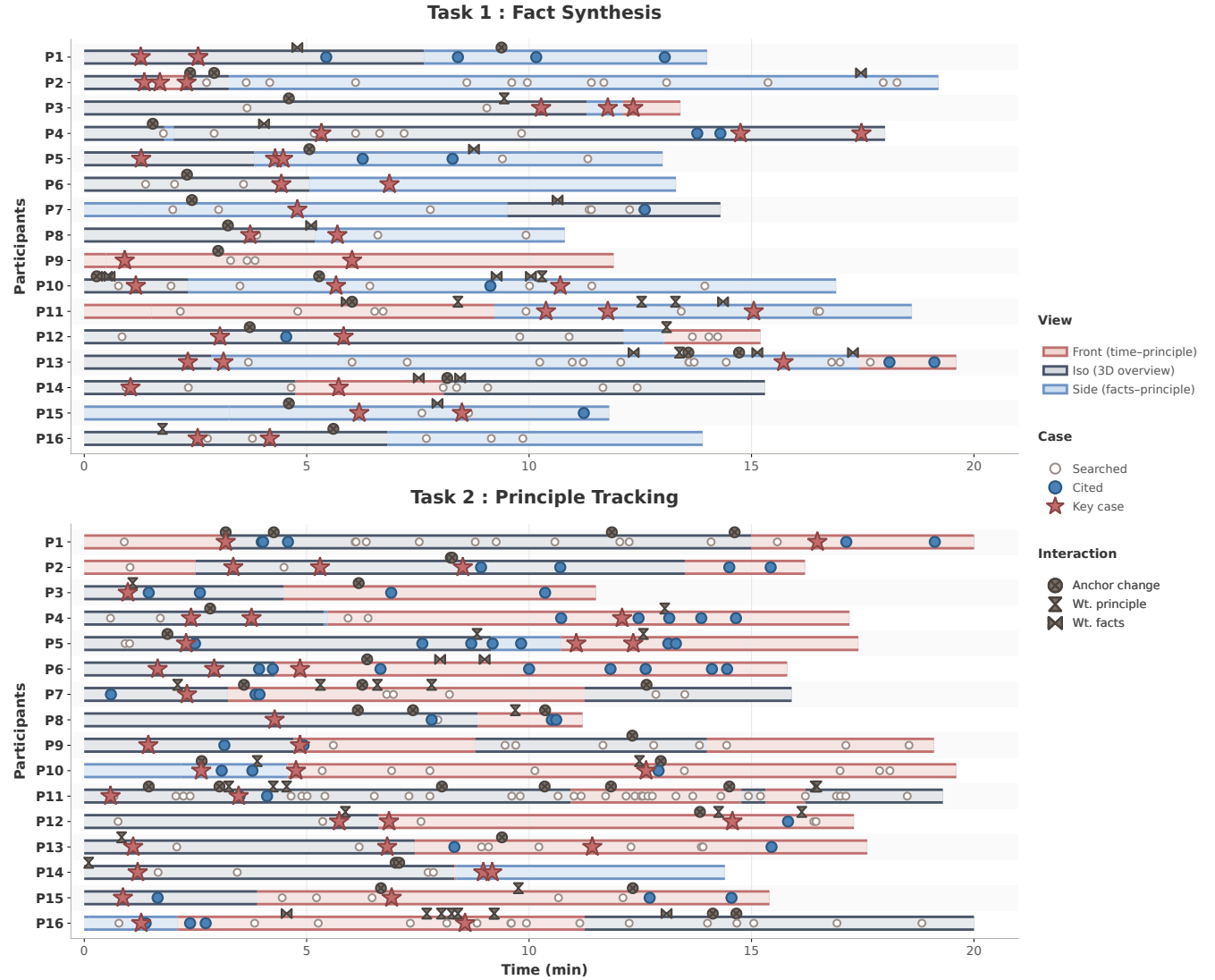


Figure 6: Interaction-annotated discovery timelines for 16 participants across both tasks. Background color encodes the active view: rose for Front (time–principle), slate for ISO (3D overview), and blue for Side (facts–principle). Markers above each bar indicate interaction events (anchor changes, weight adjustments); markers on the bar denote the case events defined in §5 (hollow circles for searched, filled blue for cited, red stars for key cases). Fact Synthesis sessions are dominated by blue (Side View) backgrounds, while Principle Tracking sessions are dominated by rose (Front View) backgrounds, corroborating the task-driven projection selection reported in §6.3.